



DESIGN AND FABRICATION OF THREE BLADED GIROMILL WIND TURBINE

A PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

SARDAR RAJA COLLEGE OF ENGINEERING,

ALANGULAM

ANNA UNIVERSITY: CHENNAI 600025

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BONAFIDE CERTIFICATE

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EXTERNAL EXAMINER

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We have immense pleasure in submitting our project. We are confident that under the able guidance and administration advice of our honorable chairman **Er. A. JESUS RAJA**, we are able to present this report successfully. We are happy in thanking him.

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ABSTRACT

This project deals with the design and fabrication of three blade Giromill wind turbine. The Giromill wind turbine is a type of vertical axis wind turbine which is used to produce power. The turbine consists of three straight blades which is technically an airfoil which is connected to the rotating main shaft. In this project the components required for this wind turbine like airfoil, main shaft and bearing are designed properly. The power calculation with respect to the velocity of wind is included. The components are fabricated with appropriate materials and assembled. Finally this project was tested and implemented successfully.

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CHAPTER 1

INTRODUCTION

The function of vertical axis wind turbine is converted the wind power to electricity. Particularly this type of turbine has the main rotor shaft arranged vertically. For it a power study is made in which the size of the wind turbine is defined.

Nowadays there are several types of vertical axis wind turbine: Savonius, Darrieus and Giromill.

The Savonius turbine is one of the simplest turbines. Looking down on the rotor from above, a two-scoop machine would look like a "S" shape in cross section. Savonius turbines are used whenever cost is much more important than efficiency. The Darrieus turbine consists of a number of curved and vertical airfoils which are attached to the central mast by horizontal supports. The advantages of variable pitch are: high starting torque; a wide, relatively flat torque curve; a lower blade speed ratio; a higher coefficient of performance; more efficient operation in turbulent winds; and a lower blade speed ratio which lowers blade bending stresses.

The Giromill turbine is a subtype of Darrieus turbine, this turbine has a straight blades. This turbine is the type of wind turbine that I am designing. It is typically formed by two or three vertical airfoils but these aren't the only options. Giromill turbine is cheaper and easier to build than a standard Darrieus turbine, but also requires strong winds (or a motor) to start. However, they work well in

turbulent wind conditions and are an affordable option where a standard horizontal axis windmill type turbine is unsuitable.

One advantage of this arrangement is that the turbine does not need to be pointed into the wind to be effective. The Giromill blade design is much simpler to build, but results in a more massive structure than the traditional arrangement, and requires stronger blades, for reasons outlined above.

CHAPTER 2

LITERATURE REVIEW

2.1. GIROMILL ROTOR

2.1.1. HISTORICAL BACKGROUND

The straight-bladed wind turbine, also named Giromill or H-rotor, is a type of vertical axis wind turbine developed by Georges Darrieus in 1927. This kind of VAWT has been studied by the Musgrove's research team in the United Kingdom during the '80.

In these turbines the “egg beater” blades of the common Darrieus are replaced with straight vertical blade sections attached to the central tower with horizontal supports. These turbines usually have 2 or 3 vertical airfoils. The Giromill blade design is much simpler to build, but results in a more massive structure than the traditional arrangement and requires stronger blades. In these turbines the generator is located at the bottom of the tower and so it can be heavier and bigger than a common generator of a HAWT and the tower can have a lighter structure.

While it is cheaper and easier to build than a standard Darrieus turbine, the Giromill is less efficient and requires motors to start. However these turbines work well in turbulent wind conditions and represent a good option in those areas where a HAWT is unsuitable.

2.1.2. PARTS OF THE ROTOR

The structure of all Giromill rotors can be splitted in three components:

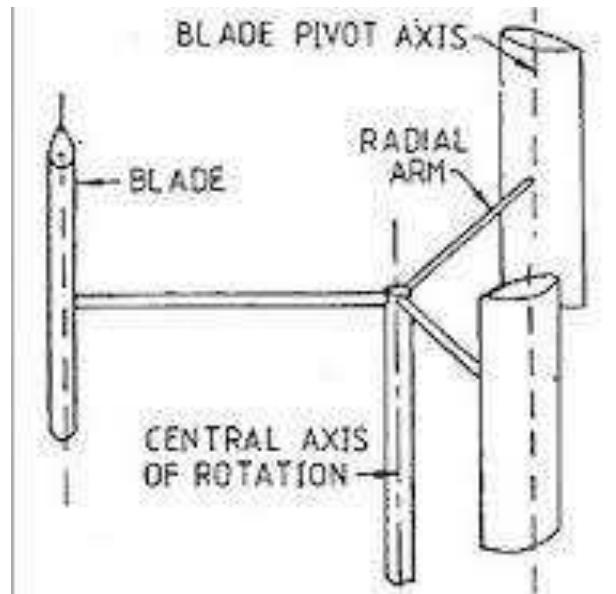


Fig. 2.1 Parts of the Rotor

- **Central axis of rotation:** This component is also called central mast. This is in the middle of the rotor. The other components of the rotor rotate around it. Generally it is metallic and its section is circular.
- **Blades:** In Giromill rotors the blades are vertical airfoils like in the Darrieus rotors, but in Giromill rotors the blades are straight. The main function of the blades is spin by the action of the wind. The blades are fixed and are connected to the rotor shaft, which rotates with them in the wind.

The number of blades is a significant issue. Generally it is a decision between two and three blades, but these aren't the only options. These machines

may suffer some efficiency loss, but adding more blades means that the machine is dealing with more drag. The VAWTs with large numbers of blade define rotation speeds relatively low. The torque ripple is reduced by using three or more blades which results in a higher solidity for the rotor in Darrieus wind turbine. The disadvantage to adding more blades is the cost.

- **Radial arms:** These components are horizontal supports which attach the blades to the central axis of rotation. It can be attached to the blades and to the central mast by welding or by means of a screwed union.

2.1.3. HOW DARRIEUS ROTOR WORKS

The working principle of Darrieus rotor can be simplified as below. First, assume that the retarded wind in front of the rotor still remains straight. When the blades are moving much faster compare to the original undisturbed wind speed i.e. ratio of blade speed to free stream wind speed, $TSR > 3$, the picture above shows the velocity vector of the airfoil blades at different angular position. With such a high TSR, the airfoils will be ‘cutting through’ the wind with small angle of attack. The resulting lift force always assists the rotor rotation while the drag force always opposes the rotation. As the lift zeroing at the left side (0 degree) and right side (180 degree) where the symmetrical airfoil moves paralleled to wind, the torque changes to negative around these position. At the near front (90 degree) and far back (270 degree) position, the lift component is much higher than the drag component, so positive torque is produced. The total torque per revolution will be positive with a good set of airfoils so the rotor will accelerate at the right direction.

During start-up, the starting torque depends on the angular position of rotor with respect to the wind direction, so the rotor might rotate at the right direction straight away or wobble a bit before starting. Normally, the rotor will need some form of assistance to reach higher revolutions before it begins to rotate by itself as the Darrieus rotor has very low torque at low TSR which can be easily worsened (till negative) by friction in the system.

To summarize the operation way of a Giromill VAWT is not different from that of a common Darrieus turbine. The wind hits the blades and its velocity is split in lift and drag component. The resultant vector sum of these two components of the velocity makes the turbine rotate. The swept area of a Giromill wind turbine is given by the length of the blades multiplied for the rotor diameter.

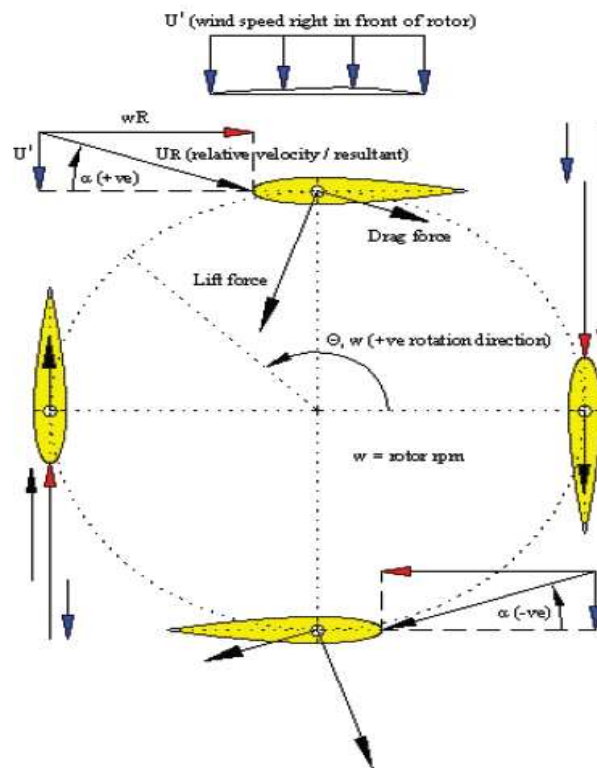


Fig. 2.2 Principle of Darrieus Rotor

One problem with the design is that the angle of attack changes as the turbine spins, so each blade generates its maximum torque at two points on its cycle (front and back of the to a sinusoidal power cycle that complicates design. Another problem arises because the majority of the mass of the rotating mechanism is at the periphery rather than at the hub, as it is with a propeller. This leads to very high centrifugal stresses on the mechanism, which must be stronger and heavier than otherwise to withstand them. The most common shape is the one similar to an eggbeater, which can avoid in part this problem, having most of the rotating mass not far from the axis.

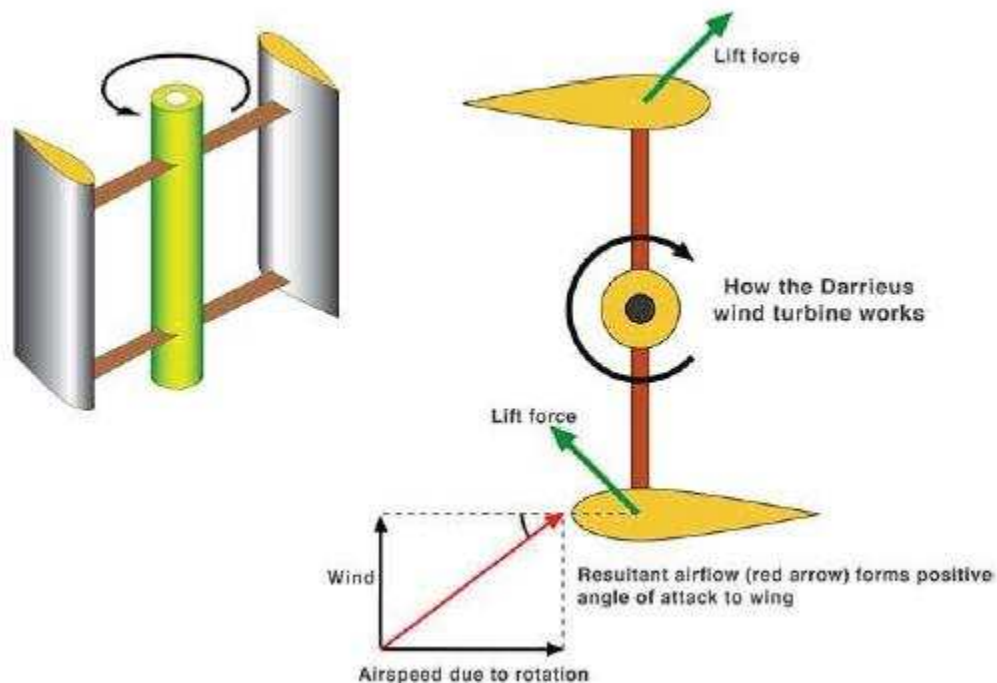


Fig. 2.3 Rotational direction of the rotor

2.2. THE EFFECT OF NUMBER OF BLADES

The number of blades is a significant issue. Generally it is a decision between two and three blades, but these aren't the only options. The benefit of adding more blades depends on a tradeoff between the various required performance characteristics and manufacturing costs.

These machines may suffer some efficiency loss, but adding more blades means that the machine is dealing with more drag. The VAWTs with large numbers of blade define rotation speeds relatively low. The torque ripple is reduced by using three or more blades which results in a higher solidity for the rotor in Darrieus wind turbine

The disadvantage to adding more blades is the cost.

Other parameters that may be affected by the number of blades include, rotor weight and balance, structural loading, torque ripple, starting torque, and fatigue resistance. It should be noted here that the number of blades was increased, but the chord of those blades was not reduced so the solidity rises substantially.

2.3. THEORY OF AERODYNAMICS

From an aerodynamic point of view, the different VAWT, have a number of aspects in common that distinguish them from the HAWT.

The blades of a VAWT rotate on a rotational surface whose axis is at right angle to the wind direction. The aerodynamic angle of attack of the blades varies

constantly during the rotation. Moreover, one blade moves on the downwind side of the other blade in the range of 180° to 360° of rotational angle so that the wind speed in this area is already reduced due to the energy extracted by the upwind blades. Hence, power generation is less in the downwind sector of rotation. Consideration of the flow velocities and aerodynamic forces shows that, nevertheless, a torque is produced in this way which is caused by the lift forces. The breaking torque of the drag forces is much lower, by comparison.

In one revolution, a single rotor blade generates a mean positive torque but there are also short sections with negative torque. The calculated variation of the total torque also shows the reduction in positive torque on the downwind side.

The alternation of the torque with the revolution can be balanced with three rotor blades, to such an extent that the alternating variation becomes an increasing and decreasing torque which is positive throughout. However, torque can only develop in a vertical axis rotor if there is circumferential speed. The vertical axis rotor is usually not self-starting.

The qualitative discussion of the flow conditions at the vertical axis rotor shows that the mathematical treatment must be more complex than with propeller type. This means that the range of physical and mathematical models for calculating the generation of power and the loading is also wider.

Various approaches, with a variety of weightings of the parameters involved have been published in the literature. Most authors specify values of 0.40 to 0.42 for the maximum C_p for the Darrieus type wind turbine.

In order to analyze the aerodynamics of a rotor and to get information about its power generation, it's necessary to start by considering that a wind turbine works converting the kinetic energy of a wind flow in electricity, following several steps:

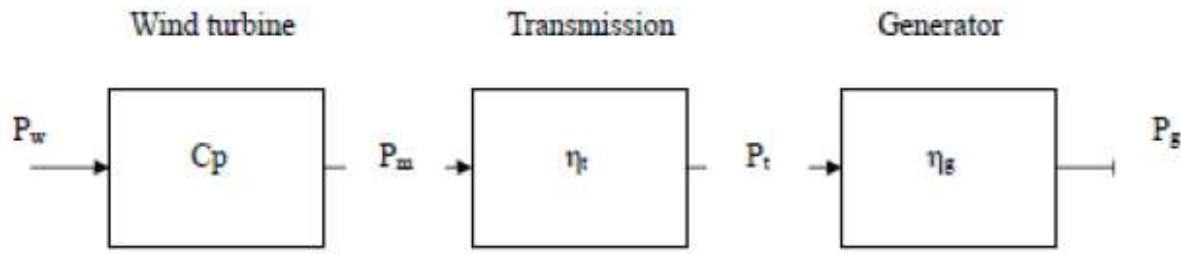


Fig. 2.4 Process of Energy Conversion

From the wind flow the turbine gets the energy to rotate the blades. The energy produced by this rotations is given to the main shaft (or to a gearbox, if it is present) and from here to the electrical generator, that provide the electricity to the grid.

2.4. POWER IN THE WIND

Another point to consider during the design of any wind turbine is how calculate the wind power in the blades. But the available wind power is different from the usable wind power. Firstly I show the expression to calculate the available wind power:

$$P = GAV^3 \quad \text{..... (2.1)}$$

Where,

P = Wind power in watts

G = Air density in kg/m³

A = area perpendicular to the wind direction formed by the rotor in m²

V = wind speed in m/s

The expression to calculate the usable wind power is:

$$P = \frac{1}{2} G A V^3 C_p \quad \dots\dots\dots (2.2)$$

In this expression C_p is the power coefficient that depends on the type of machine and for each variable in turn to the relationship between the peripheral speed of the blades and wind speed.

To calculate the wind speed exist this expression:

$$V = V_0 \left(\frac{h}{h_0} \right)^n \quad \dots\dots\dots (2.3)$$

V is the wind speed at height h above ground in m/s

V_0 is the known wind speed at a height h_0 m/s

h is the height at which the wind speed to be calculated in m

h_0 is the reference height in m

n is the value which depends on the terrain roughness:

- Smooth (sea, sand, snow): 0.10-0.13
- Moderately rugose: 0.13-0.20
- Rugose (forest, neighborhoods): 0.20-0.27
- Very rugose (cities, tall buildings): 0.27-0.40

2.5. POWER COEFFICIENT

The power output of a wind turbine rotor changes with the rpm so the rotor performance is normally presented in power coefficient versus tip to wind speed

ratio graph. Sometimes, the same information is also expressed in form of torque coefficient vs. tip speed ratio curve. For a Darrieus wind turbine:

The power coefficient is defined as:

$$C_p = \frac{P}{\frac{1}{2}\rho U^3 A} \dots\dots\dots (2.4)$$

Where,

P = rotor power

The tip to wind speed ratio, or tip speed ratio, or TSR is defined as:

$$\lambda = \frac{wR}{U} \dots\dots\dots (2.5)$$

Where,

R = maximum rotor radius

w = rotor rpm

The torque coefficient is defined as:

$$C_t = \frac{T}{\frac{1}{2}\rho U^3 A} \dots\dots\dots (2.6)$$

Where,

T = rotor torque

The relation between these two coefficients is:

$$C_t = \frac{C_p}{\lambda} \dots\dots\dots (2.7)$$

3.6. BETZ'S LAW

Betz's law is a theory about the maximum possible energy to be derived from a "hydraulic wind engine", or a wind turbine such as the Éolienne Bollée (patented in 1868), the Eclipse Windmill (developed in 1867), and the Aermotor (first appeared in 1888 to pump water for cattle, and is still in production). Decades before the advent of the modern 3-blade wind turbine that generates electricity, Betz's law was developed in 1919 by the German physicist Albert Betz. According to Betz's law, no turbine can capture more than 59.3 percent of the kinetic energy in wind. The ideal or maximum theoretical efficiency $\eta_{\max}$ (also called power coefficient) of a wind turbine is the ratio of maximum power obtained from the wind to the total power available in the wind. The factor 0.593 is known as Betz's coefficient (from the name of the man who first derived it). It is the maximum fraction of the power in a wind stream that can be extracted. The power coefficient is defined as:

$$\text{Power coefficient, } C_p = \frac{\text{Power output from wind machine}}{\text{Power available in wind}} \dots\dots\dots (2.8)$$

CHAPTER 3

DESIGN OF GIROMILL WIND TURBINE

3.1 INTRODUCTION

The experimental setup was hand sketched for initialization of the components required and about their dimensions which is shown in Fig. 3.1.

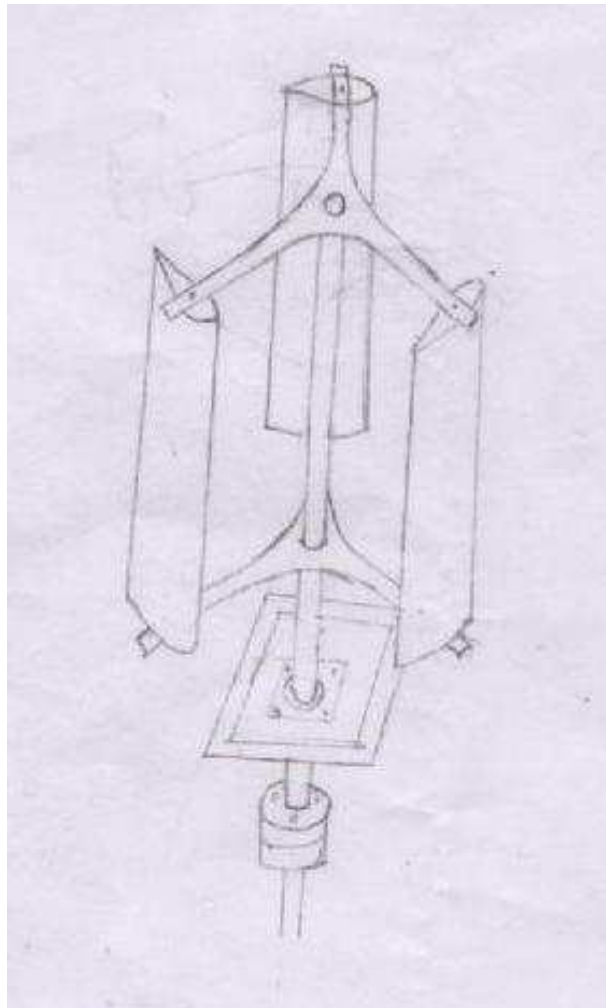


Fig. 3.1 Hand Sketch of Setup

The main components of the Giromill wind turbine are:

1. Airfoil
2. Radial arms
3. Shaft
4. Bearings
5. Screws
6. Support

3.2 AIRFOIL

The type of airfoils that I have studied to apply in my wind turbine is a four-digit NACA wing section. In this type of NACAs airfoil the first digit describing maximum camber as percentage of the chord, the second digit describing the distance of maximum camber from the airfoil leading edge in tens of percents of the chord and the last two describing maximum thickness of the airfoil as percent of the chord.

Particularly the NACAs airfoil that we have analyzed in my project is symmetrical 4- digit NACAs airfoil. The main characteristic of these NACAs is that their first two digits are 0. The formula for the shape of a NACA 00xx foil, with "xx" being replaced by the percentage of thickness to chord, is:

$$y = \frac{y}{0.2} c \left[0.2969 \sqrt{\frac{x}{c}} - 0.1260 \left(\frac{x}{c} \right) - 0.3516 \left(\frac{x}{c} \right)^2 + 0.2843 \left(\frac{x}{c} \right)^3 - 0.1015 \left(\frac{x}{c} \right)^4 \right]$$

..... (3.1)

Where,

c is the chord length

x is the position along the chord from 0 to c

y is the half thickness at a given value of x (centerline to surface)

t is the maximum thickness as a fraction of the chord (so 100 t gives the last two digits in the NACA 4-digit denomination)

This formula will be very important to draw the airfoils in hand or using softwares like Gambit by the coordinates generated by one generator of points.

Exactly it is analyzed the behavior in 2D of these three symmetrical NACA's airfoils: NACA 0012, NACA 0015 and NACA 0020. These three NACA's were chosen because they have given good results in previous studies carried out about Giromill turbines.

Out of this it is selected the NACA 0015 for this project.

The table 3.1 shows the coordinates of the NACA 0015:

Table 3.1 Coordinates of NACA 0015 airfoil

X	Y(UPPER)	Y(LOWER)
1	0.00221	0.00221
0.95	0.01412	0.01412
0.9	0.02534	0.02534
0.8	0.04591	0.04591
0.7	0.06412	0.06412
0.6	0.07986	0.07986
0.5	0.09265	0.09265
0.4	0.10156	0.10156
0.3	0.10504	0.10504
0.25	0.10397	0.10397
0.2	0.1004	0.1004
0.15	0.09354	0.09354
0.1	0.08195	0.08195
0.075	0.0735	0.0735
0.05	0.06221	0.06221
0.025	0.04576	0.04576
0.0125	0.03315	0.03315
0	0	0

The Fig. 3.2 shows the hand sketching of the NACA 0015 airfoil:

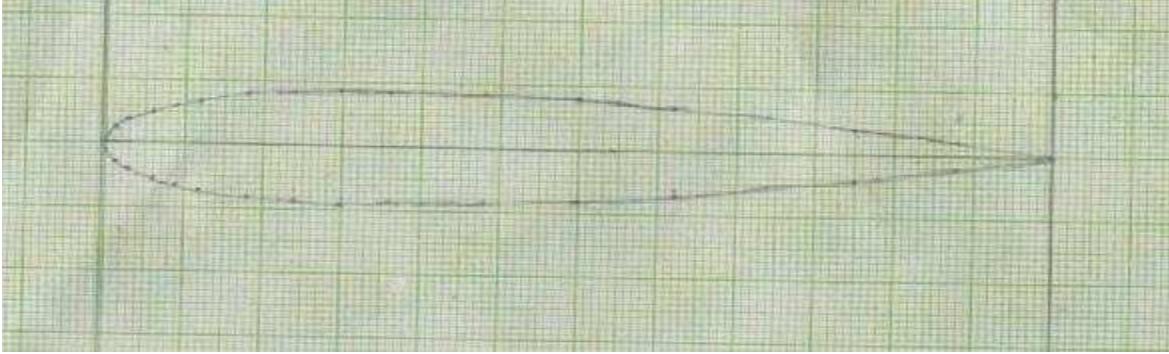


Fig. 3.2 Hand sketching of NACA 0015 airfoil

The airfoil has a hole in top and bottom surface to connect with the radial connecting arms. This can be clearly seen in the Fig. 3.3 which is generated using the Pro-E software. The place of the drill is determined as 40% of chord from the front end based on the previous studies of the Giromill wind turbine.

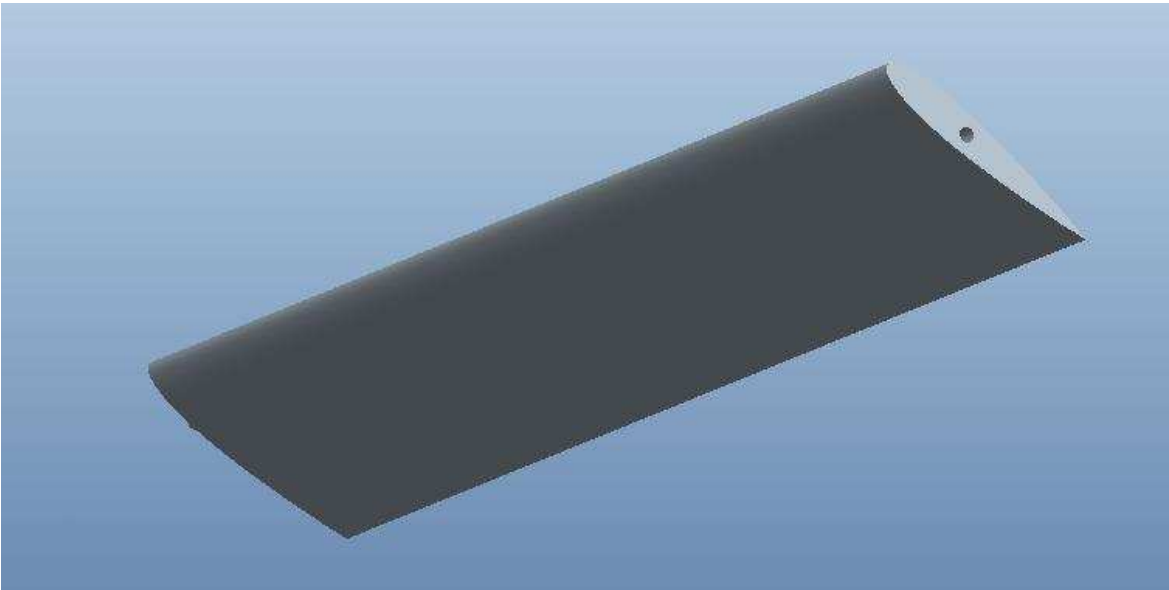


Fig. 3.3 3D model of the blade

3.3 RADIAL ARMS

The radial arms are the components that are used to connect the air foil to centre shaft. Specific systems are needed to be installed for easy assembling and dismantling of airfoil and shaft. The material used for fabrication of radial arms is plywood because of its light weight and high strength.

To fasten the airfoil with the arms 6mm screws are used and also for fastening the centre shaft. There two types of the radial arms is designed one for the top and another for the bottom which goes through the shaft.

TOP RADIAL ARM DIMENSIONS

Table 3.2 Dimensions of top radial arm

SNO.	DESCRIPTION	DIMENSION
1	Number of arms	3
2	Total arm length	20cm
3	Centre disc diameter	5cm
4	Centre drill diameter	0.6cm
5	Side holes diameter	0.6cm
6	Side holes position from the outer end	1.5cm
7	Width of the arm	3cm
8	Thickness	0.9cm

The Fig. 3.4 shows the 3d model of the top radial arms which is generated using the Pro- e software:

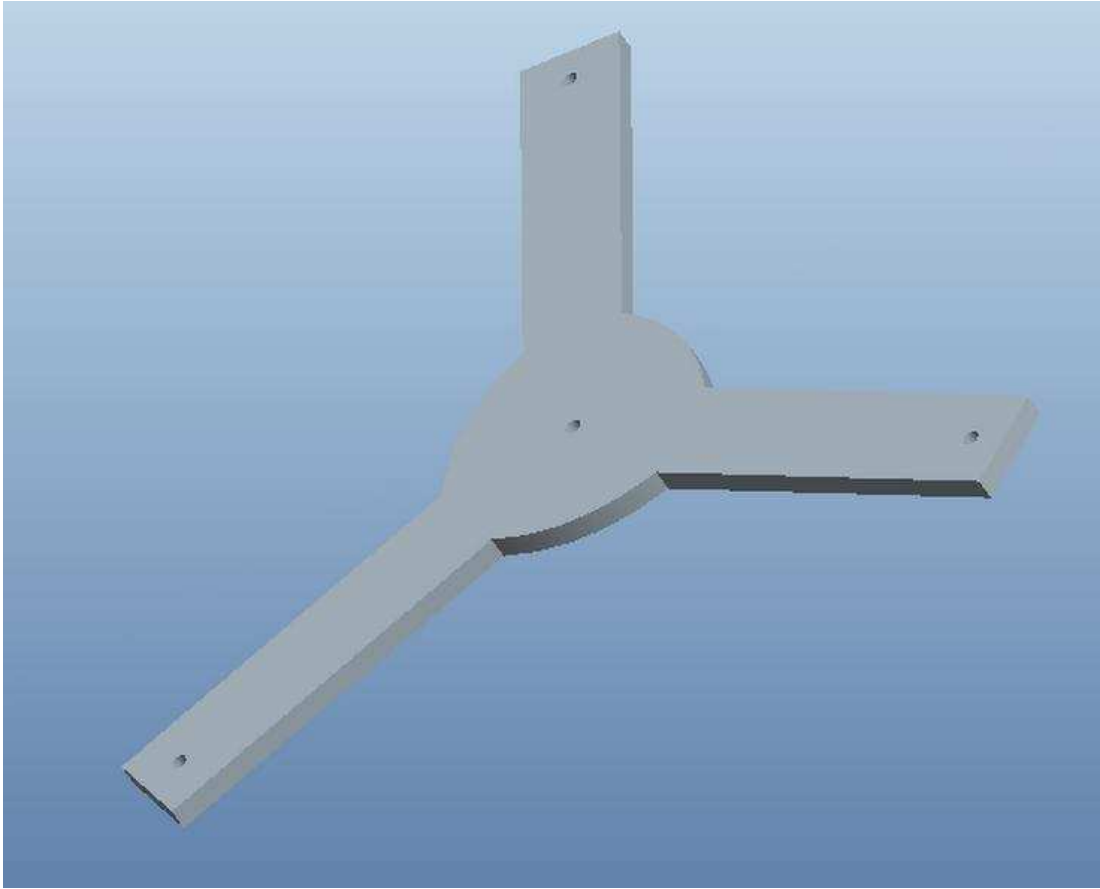


Fig. 3.4 Top radial arm

The bottom radial arm has the same design and dimensions except the centre hole is 16mm rather than 6mm.

BOTTOM RADIAL ARM DIMENSIONS

Table 3.3 Dimensions of bottom radial arm

SNO.	DESCRIPTION	DIMENSION
1	Number of arms	3
2	Total arm length	20cm
3	Centre disc diameter	5cm
4	Centre drill diameter	1.6cm
5	Side holes diameter	0.6cm
6	Side holes position from the outer end	1.5cm
7	Width of the arm	3cm
8	Thickness	0.9cm

The Fig. 3.5 shows the 3d model of the bottom radial arm which is generated using the Pro- e software:

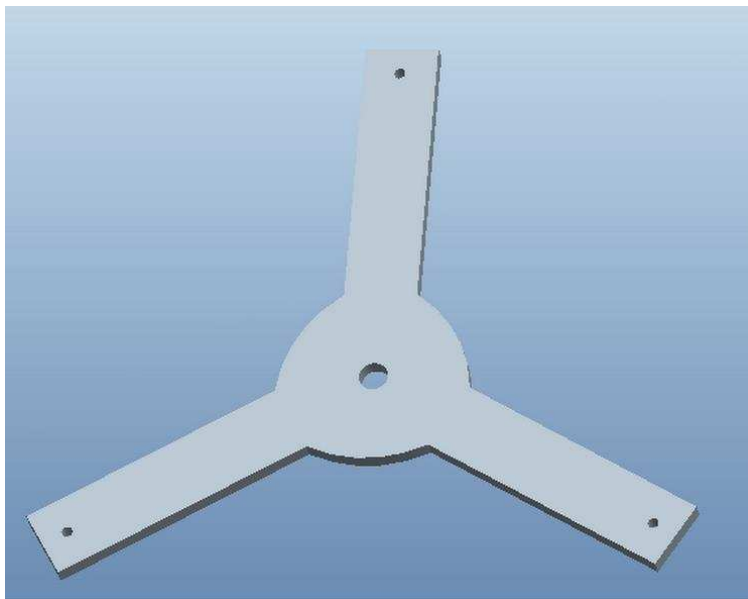


Fig. 3.5 Bottom radial arm

3.4 SHAFT

Shaft is a rotating machine element which is used to transmit power from one place to another. The power is delivered to the shaft by some tangential force and the resultant torque set up within the shaft permits the power to be transferred to various machines linked up to the shaft. In order to transfer the power from one shaft to another, the various members such as pulleys, gears etc. are mounted on it. These members along with the forces exerted upon them cause the shaft to be used for the transmission of torque and bending moment. In other words, one may say that a shaft is used for the transmission of torque and bending moment. The various members are mounted on the shaft by means of keys or splines.

The shaft is usually cylindrical, but may be square or cross-shaped in section. They are solid in cross-section but sometimes hollow shafts are also used. An axle, though similar in shape to the shaft, is a stationary machine element and is used for the transmission of bending moment only. It simply acts as a support for some rotating body such as a hoisting drum, a car wheel or a rope. A spindle is a short shaft that imparts motion.

3.4.1 MATERIALS USED FOR SHAFT

The used for shafts should have the following properties:

1. It should have high strength.
2. It should have good machinability.
3. It should have low notch sensitivity factor.
4. It should have good heat treatment properties.
5. It should have high wear resistant properties.

Based on the calculations the shaft diameter is determined as 16mm. to reduce the weight of the shaft the material is selected as aluminum due to its low weight and high strength properties that highly suits the requirements of the wind turbine.

A 6mm hole with thread in it to be drilled on the top flat surface of the shaft. The Table 3.4 shows the dimensions of the shaft:

Table 3.4 Dimensions of the shaft

SNO.	DESCRIPTION	DIMENSION
1	Diameter of shaft	1.6cm
2	Length of shaft	100cm
3	Diameter of hole	0.6cm

The Fig. 3.6 shows the 3d model of the shaft which is generated using the Pro-E software.



Fig. 3.6 3d model of the shaft

3.5 BEARINGS

Minimizing required start-up torque is essential for the wind turbine to self-start. Without proper bearings our wind turbine will either not operate properly, or ruin the bearings that were used improperly, which could result in unsafe operating conditions. Bearings can be very expensive, and for our particular setup we will require 2 roller bearings that are going to primarily centralize the shaft, and a turntable bearing to take the majority of the weight. This combination will provide

the least amount of friction, while maximizing bearing life and maintaining safe operating conditions.



Fig. 3.7 Bearing

3.6 SCREWS

A screw, or bolt, is a type of fastener characterized by a helical ridge, known as an external thread or just thread, wrapped around a cylinder. Some screw threads are designed to mate with a complementary thread, known as an internal thread, often in the form of a nut or an object that has the internal thread formed into it. Other screw threads are designed to cut a helical groove in a softer material as the screw is inserted. The most common uses of screws are to hold objects together and to position objects.

A screw will almost always have a head, (a set screw is an example of a screw without a head) which is a specially formed section on one end of the screw that allows it to be turned, or driven. Common tools for driving screws include

screw drivers and wrenches. The head is usually larger than the body of the screw, which keeps the screw from being driven deeper than the length of the screw and to provide a bearing surface. There are exceptions; for instance, carriage bolts have a domed head that is not designed to be driven; set screws often have a head smaller than the outer diameter of the screw; J-bolts have a J-shaped head which is not designed to be driven, but rather is usually sunk into concrete allowing it to be used as an anchor bolt. The cylindrical portion of the screw from the underside of the head to the tip is known as the shank; it may be fully threaded or partially threaded.



Fig. 3.8 Screw

3.7 SUPPORT

For most of the mechanical equipment balancing is very necessary, while changing into different positions there should be given more importance of stableness or balancing. To achieve this supports are required. As it in the vertical axis wind turbine it must compensate the cyclic forces produced due to the varying torque of the blades in every rotation. The continuous operation of the wind turbine causes fatigue stresses in the support elements. This problem must be eliminated by choosing the material which has high fatigue strength.

CHAPTER 4

DESIGN CALCULATIONS

4.1 AIRFOIL CALCULATIONS

As per the equation 3.1, for finding the coordinates of the airfoil the following formula must be taken into account.

$$y = \frac{t}{0.2} c \left[0.2969 \sqrt{\frac{x}{c}} - 0.1260 \left(\frac{x}{c} \right) - 0.3516 \left(\frac{x}{c} \right)^2 + 0.2843 \left(\frac{x}{c} \right)^3 - 0.1015 \left(\frac{x}{c} \right)^4 \right] \dots\dots\dots (3.1)$$

For NACA 0015,

Maximum thickness, t : 15% of chord length

Here,

Chord length, c : 12cm

Therefore,

Maximum thickness = 12x0.15

Maximum thickness = 1.8cm

Here the position along the chord (x) varies from 0 to 12.

The position of the chord is initiated as follows,

0.000, 0.150, 0.300, 0.600, 0.900, 1.200, 1.800, 2.400, 3.000,
3.600, 4.800, 6.000, 7.200, 8.400, 9.600, 10.800, 11.400, 12.000.

At $x_1=0\text{cm}$,

$$y_1 = \frac{t}{0.2} c \left[0.2969 \sqrt{\frac{x_1}{c}} - 0.1260 \left(\frac{x_1}{c} \right) - 0.3516 \left(\frac{x_1}{c} \right)^2 + 0.2843 \left(\frac{x_1}{c} \right)^3 - 0.1015 \left(\frac{x_1}{c} \right)^4 \right]$$

$$y_1 = \frac{t}{0.2} 12 \left[0.2969 \sqrt{\frac{0}{c}} - 0.1260 \left(\frac{0}{c} \right) - 0.3516 \left(\frac{0}{c} \right)^2 + 0.2843 \left(\frac{0}{c} \right)^3 - 0.1015 \left(\frac{0}{c} \right)^4 \right]$$

$$y_1 = 0.000$$

At $x_2=0.150\text{cm}$,

$$y_2 = \frac{t}{0.2} c \left[0.2969 \sqrt{\frac{x_2}{c}} - 0.1260 \left(\frac{x_2}{c} \right) - 0.3516 \left(\frac{x_2}{c} \right)^2 + 0.2843 \left(\frac{x_2}{c} \right)^3 - 0.1015 \left(\frac{x_2}{c} \right)^4 \right]$$

$$y_2 = \frac{t}{0.2} 12 \left[0.2969 \sqrt{\frac{0.15}{c}} - 0.1260 \left(\frac{0.15}{c} \right) - 0.3516 \left(\frac{0.15}{c} \right)^2 + 0.2843 \left(\frac{0.15}{c} \right)^3 - 0.1015 \left(\frac{0.15}{c} \right)^4 \right]$$

$$y_2 = 0.01896\text{cm}$$

At $x_3=0.300\text{cm}$,

$$y_3 = \frac{t}{0.2} c \left[0.2969 \sqrt{\frac{x_3}{c}} - 0.1260 \left(\frac{x_3}{c} \right) - 0.3516 \left(\frac{x_3}{c} \right)^2 + 0.2843 \left(\frac{x_3}{c} \right)^3 - 0.1015 \left(\frac{x_3}{c} \right)^4 \right]$$

$$y_3 = \frac{t}{0.2} 12 \left[0.2969 \sqrt{\frac{0.3}{c}} - 0.1260 \left(\frac{0.3}{c} \right) - 0.3516 \left(\frac{0.3}{c} \right)^2 + 0.2843 \left(\frac{0.3}{c} \right)^3 - 0.1015 \left(\frac{0.3}{c} \right)^4 \right]$$

$$y_3 = 0.39456\text{cm}$$

At $x_{17}=11.400\text{cm}$,

$$y_{17} = \frac{t}{0.2} c \left[0.2969 \sqrt{\frac{x_{17}}{c}} - 0.1260 \left(\frac{x_{17}}{c} \right) - 0.3516 \left(\frac{x_{17}}{c} \right)^2 + 0.2843 \left(\frac{x_{17}}{c} \right)^3 - 0.1015 \left(\frac{x_{17}}{c} \right)^4 \right]$$

$$y_{17} = \frac{t}{0.2} 12 \left[0.2969 \sqrt{\frac{11.400}{c}} - 0.1260 \left(\frac{11.400}{c} \right) - 0.3516 \left(\frac{11.400}{c} \right)^2 + 0.2843 \left(\frac{11.400}{c} \right)^3 - 0.1015 \left(\frac{11.400}{c} \right)^4 \right]$$

$$y_{17} = 0.12096\text{cm}$$

The calculations are done for all the 18 x values and the results are plotted in table 4.1.

Table 4.1 Airfoil coordinates

X	Y(UPPER)	Y(LOWER)
12	001896	001896
11.4	0.12096	0.12096
10.8	0.2232	0.2232
9.6	0.39348	0.39348
8.4	0.5496	0.5496
7.2	0.68448	0.68448
6	0.79404	0.79404
4.8	0.87048	0.87048
3.6	0.87624	0.87624
3	0.89124	0.89124
2.4	0.86064	0.86064
1.8	0.80184	0.80184
1.2	0.69996	0.69996
0.9	0.0.63	0.0.63
0.6	0.53316	0.53316
0.3	0.39456	0.39456
0.15	0.28404	0.28404
0	0	0

4.2 DESIGN OF SHAFT

$$P = \frac{2\pi NT}{60} \dots\dots\dots (4.1)$$

Torque

$$T = Wr \dots\dots\dots (4.2)$$

Where,

$$\text{Radius, } r = 18.5\text{cm} = 185\text{mm}$$

$$W = \text{Sum of weights of airfoils}$$

$$\text{Load, } W = W_1 + W_2 + W_3 \dots\dots\dots (4.3)$$

Here the weights of the airfoils are equal,

$$\text{Load, } W = 3W_1 \dots\dots\dots (4.4)$$

Here,

Weight of airfoil 1, $W_1 = 500\text{gms} = 0.5\text{kg}$

$$W_1 = 0.5 \times 9.81$$

$$W_1 = 4.905\text{N}$$

$$W = 3 \times 4.905$$

$$\text{Load, } W = 14.715\text{N}$$

The weight of the radial arm is considered to be negligible.

$$T = Wr$$

$$T = 14.715 \times 185$$

$$T = 2722.275 \text{ Nmm} = 2.72 \text{ Nm}$$

MODEL INPUTS

Table 4.2 Model inputs

S.NO	WIND SPEEDS(m/s)	SPEED(RPM)
1	2	8
2	5	14
3	7	25
4	10	36
5	15	49
6	20	70

Taking $N=70$ rpm

$$\text{Power, } P = \frac{2 * \pi * 70 * 2.72275}{60}$$

$$P = 19.96 \text{ KW}$$

$$\text{Power, } P = 20 \text{ KW}$$

Shear stress acting on the shaft

$$T = \frac{\pi}{16} \tau d^3 \quad \dots\dots\dots (4.5)$$

$$2.72275 = \frac{\pi}{16} \tau * 0.016^3$$

$$\tau = 3.385 \times 10^6 \text{ (N/m}^2\text{)}$$

Compressive stress acts on the shaft

$$\sigma_c = \frac{P_c}{A} \dots\dots\dots (4.6)$$

$$P_c = (\text{weight of 3 airfoils} + \text{weight of 2 radial arms}) \dots\dots\dots (4.7)$$

$$P_c = (1.5 + 0.2) * 9.81$$

$$P_c = 149.112N$$

Cross-sectional area of shaft, $A = \frac{\pi}{4} d^2 \dots\dots\dots (4.8)$

$$A = \frac{\pi}{4} * 0.016^2$$

$$A = 2.0106 \times 10^{-4} (m^2)$$

$$\sigma_c = \frac{149.112}{2.0106 \times 10^{-4}}$$

$$\text{Compressive stress, } \sigma_c = 0.7416 \times 10^6 (N/m^2)$$

Here the design stresses of Aluminium are:

$$\text{Tensile yield strength} = 276 \times 10^6 (N/m^2)$$

$$\text{Shear yield strength} = \frac{\text{Tensile yield strength}}{2} \dots\dots\dots (4.9)$$

$$= \frac{276 \times 10^6}{2}$$

$$\text{Shear yield strength} = 138 \times 10^6 (N/m^2)$$

$$\text{Design shear stress, } [\tau] = \frac{\text{Shear yield strength}}{\text{Factor of safety}} \dots\dots\dots (4.10)$$

Assuming, Factor of safety = 4

$$[\tau] = \frac{138 \times 10^6}{4}$$

$$\text{Design shear stress, } [\tau] = 34.5 \times 10^6 \text{ (N/m}^2\text{)}$$

On comparing,

$$(34.5 \times 10^6 > 3.385 \times 10^6 | [\tau] > \tau)$$

Therefore, design is safe.

4.3 RADIAL ARMS CALCULATION

The arms have the bending load in it.

On observing its section it is founded that it has a cantilever beam like structure with eccentric loading. In which the load is the weight of the airfoil.

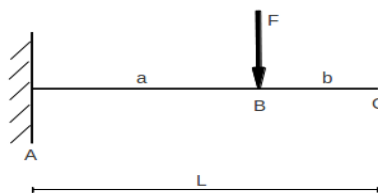


Fig. 4.1 Bending load on radial arms

Here,

$$L = 0.15\text{m}$$

$$a = 0.135\text{m}$$

$$b = 0.015\text{m}$$

$$F = 0.5\text{kg} = 4.905\text{N}$$

$$E = 9.3 \times 10^9 \text{ N/m}^2$$

Deflection of beam at point B

$$y_B = \frac{W a^3}{3EI} \dots\dots\dots (4.11)$$

Where,

- W - Weight of the airfoil in N
- a - Distance from the support to the load in m
- b - Distance from the load to the free end in m
- E - Young's modulus in N/m^2
- I - Moment of inertia in m^4

Here,

$$E = 9.3 \times 10^9 \text{ N/m}^2$$

$$I = \frac{tw^3}{12} \quad \dots\dots\dots (4.12)$$

Where,

w - Width of arm in m = 0.02m

t - Thickness of arm in m = 0.009m

$$I = \frac{0.009 \times 0.02^3}{12}$$

$$I = 6 \times 10^{-9} \text{ m}^4$$

$$y_B = \frac{4.905 \times 0.135^3}{3 \times 9.3 \times 10^9 \times 6 \times 10^{-9}}$$

$$y_B = 7.209 \times 10^{-5} \text{ m} = 0.0721 \text{ mm}$$

Deflection of beam at point C

$$y_C = \frac{wa^3}{3EI} + \frac{wa^2}{2EI} b \quad \dots\dots\dots (4.13)$$

$$y_C = \frac{4.905 \times 0.135^3}{3 \times 9.3 \times 10^9 \times 6 \times 10^{-9}} + \frac{4.905 \times 0.135^2}{2 \times 9.3 \times 10^9 \times 6 \times 10^{-9}} \times 0.015$$

$$y_C = 8.41 \times 10^{-5} \text{ m} = 0.0841 \text{ mm}$$

4.4 DESIGN OF BEARINGS

Radial load, $F_R = [\text{weight of three airfoils, two radial arms and shaft}] \times 9.81$

..... (4.14)

$$F_R = [(0.5 \times 3) + (0.2 \times 2) + (0.6/2)] \times 9.81$$

$$F_R = 21.582 \text{ N}$$

Axial load, $F_a = [(0.5 \times 2) \times 2] + (0.3 \times 9.81)$

$$\frac{F_a}{F_r} = \frac{12.753}{21.582}$$

$$\frac{F_a}{F_r} = 0.591$$

For, Inner diameter of the shaft, $d = 17 \text{ mm}$

Dynamic capacity, $C_o = 2850 \text{ N}$

At, Bearing number, $= 6003$

$$\frac{F_a}{C_o} = \frac{12.753}{2850}$$

$$\frac{F_a}{C_o} = 0.05$$

Corresponding to $\frac{F_a}{F_r} = 0.591$ and $\frac{F_a}{C_o} = 0.05$

Radial factor, $X = 0.56$

Thrust factor, $Y = 1$

Service factor, $S = 1.5$

Equivalent load, $P = [X \cdot F_r + Y \cdot F_a] S$ (4.15)

$$P = [(0.56 \cdot 21.582) + (2 \cdot 12.753)] \cdot 1.5$$

$$P = 37.258 \text{ N}$$

Static capacity, $C = \left(\frac{L}{L_{10}} \right)^{1/k} P$ (4.16)

$$K = 3 \quad (\text{for ball bearings})$$

$$L_{10} = 1 \times 10^6 \text{ rev}$$

Life in hours, $L_H = 24 \text{ hours in } 3 \text{ years}$

$$L_H = 3 \times 365 \times 24$$

$$L_H = 26280 \text{ hrs}$$

Life in revolutions, $L = 60 N L_H$ (4.17)

$$L = 60 \times 300 \times 26280$$

$$L = 473.04 \times 10^6 \text{ rev}$$

$$C = \left(\frac{473.04 \times 10^6}{1 \times 10^6} \right)^{1/3} 37.38$$

$$C = 290.303 \text{ N} = 29.0303 \text{ kgf}$$

For 17mm of shaft diameter,

Bearing number = 6003

Inner diameter, $d = 17 \text{ mm}$

Outer diameter, $D = 35 \text{ mm}$

Width of bearing $B = 10\text{mm}$

Static loading $C_o = 285\text{kgf}$

Dynamic loading $C = 465\text{kgf}$

Here,

$$([C] > C | 465\text{kgf} > 29.03\text{kgf})$$

Therefore, the design of bearing is safe.

CHAPTER 5

FABRICATION

5.1 AIRFOIL AND RADIAL ARMS FABRICATION

The designing of three bladed Darrieus wind turbine was completed in the previous section. The next process was fabrication.

5.1.1 PURCHASE OF MATERIAL

The fabrication of airfoil and radial arm was initially started with purchasing the raw materials required. The nute wood one of the engineered woods was brought from wood store nearby Tirunelveli. A 9mm thickness of plywood is brought about the dimensions of 2 x 1 feet.

5.1.2 FABRICATION OF AIRFOIL

Since the thickness of wood sheet is 10mm which is not enough to carve the airfoil, two sheets are bonded together with the help of the Fevicol by applying it between the two sheets and they are attached together with the help of three clamps to hold it tightly. After 10 hours the sheets are unclamped and the bonding of the sheets was checked.

The next step is the carving of the wooden sheets into the shape of the airfoil. At first the bonded sheets are cut into three pieces of required length. The profile of the airfoil is plotted on the graph sheet and it is pasted over cross-section of the bonded sheets. As the drawn profile in the graph sheet as the guide the airfoil is carved.

The bonded sheets are roughly carved with the help of wood carving machine. Then it carved with hands to get a fair profile of the airfoil. Again it is perfectly finished with the help of the salt paper by rubbing over it. This final process is also used to remove small amount of material from the surface to get the accurate profile of the airfoil.

The Fig. 5.1 shows the airfoil after the fabrication process is done.



Fig. 5.1 Airfoil

5.1.3 FABRICATION OF RADIAL ARMS

The profile of the radial arm is first drawn over a chart. Using this as the guide, the plywood sheet is cut using the machine wood cutter which is generally used for cutting designs with wood sheets. Thus the two pieces of radial arms are cut as stated in the above process.



Fig. 5.2 Fabrication of radial arms

5.1.4 DRILLING PROCESS IN AIRFOIL AND RADIAL ARMS

Following the first two processes of carving and cutting, the third and final process is the drilling of the airfoil and radial arms at the required places and at necessary. The 6mm drill is made at the airfoil at 4cm from the leading edge based on the previous studies of the Giromill wind turbine.



Fig. 5.3 Radial drilling machine

At the top radial arm 6mm drill is made at centre as well as in the ends of the three arms with 1.5cm from the free end. On the other hand on the bottom radial arm the hole places are same as in the top but the drill diameter is 16mm rather than 6mm. This hole is carried out by two drill process. The first one is with the 6mm drill bit followed by the 16mm drill bit.

5.2 MACHINING OF SHAFT

Since the aluminium shafts are readily available in stores for required diameters, a 16mm diameter aluminium shaft of 1 meter is brought from the metal shop which is nearby Tirunelveli. The facing operation is done using the help of lathe followed by the drilling of 6mm on the same face.

The hole is then threaded using the tap set for the fastening of the screw with the radial arm. For power transmission a pulley drive is employed. The larger pulley is manufactured by making a groove cut in the nylon bush which is available in the hardware store.

The smaller pulley is also manufactured using the same process explained above. The bigger pulley is drilled to a hole of 16mm and it is inserted into the shaft. The smaller pulley is fitted to the generator shaft.



Fig. 5.4 Turning process

After these processes have been completed the shaft is found out to be in slight bending due to the machining operations. This misalignment is aligned using the levelling process which is done using the lathe.

The Fig. 5.5 shows the actual photo of the shaft after machining.



Fig. 5.5 Shaft

5.3 FABRICATION OF SUPPORT

5.3.1 MATERIAL SELECTION AND PURCHASE

The raw material for the fabrication of stand is selected as the 1" x1" L-Angle bar for its high strength and easy availability. These were purchased in the iron and steel company nearby Alangulam.

5.3.2 CONVERSION OF RAW MATERIALS INTO REQUIRED WORK PIECE

The L-Angle bars are cut into four pieces of length 50cm, 20cm and 52cm. The power hack saw that used to cut the steel pieces is shown in Fig 5.3.



Fig. 5.6 Power hack saw

5.3.3 WELDING

All the required cast iron L-Angles were cut into pieces according to the dimensions. The next step in the project is to assemble the required components. In this project since the support is considered to be a solid part. To attain this welding process were used. The employed welding process was called as Arc welding. The typical welding process is shown in the Fig. 5.4.



Fig. 5.7 Arc welding process

The fabricated support using the above methods is shown in the Fig. 5.5.



Fig. 5.8 Support

5.4 ASSEMBLY

The required processes were carried out to proceed to the final stage of this project. The last stage was nothing but assembly. All the assembling processes were carried out as per the modeling of the project. This assembling process is carried out into two phases. The first phase is the rotor assembly and the second stage is the shaft and support assembly. The final process is the assembling of the total setup.

5.4.1 ROTOR ASSEMBLY

The components of the rotor are the radial arms and the airfoil. The rotor can be assembled by fastening the screws of in the slots at the top and bottom surface of the airfoil in which the radial arms are in-between them. The Fig. 5.6 shows the rotor assemblage.

The Fig. 5.9 shows the photography of the rotor assembly.



Fig 5.9 Rotor assembly

5.4.2 SHAFT AND SUPPORT ASSEMBLY

This is a simplest phase of the assembling process. In this process the bearings are first inserted into the slots which are already shown in Fig. 5.5. The shaft is then inserted into the support through the bearings. The Fig. 5.7 shows the shaft and support assembly.



Fig. 5.10 Shaft and support assembly

5.4.3 TOTAL ASSEMBLY

This is the final phase of the project. In this process the rotor is first inserted through the shaft. Then the shaft and rotor is connected with the help of the 6mm screw which is fastened at the top of the radial arm. Thus the project was assembled using the above shown means and the total setup is shown in Fig. 5.8



Fig 5.11 Total assembly

CHAPTER 6

COST ESTIMATION

Table 6.1 Cost Estimation

Sl.NO	DESCRIPTION	MATERIAL	QUANTITY	AMOUNT (IN RUPEES)
1	Shaft	Aluminium	1m x 1	210
2	Bearings	Bronze	2	150
3	Wooden sheet	Nute	2 x 2ft *1	585
4	Wooden sheet	Water proof Plywood	1 x 2 ft *1	200
5	L-Angle 20x20mm	Cast iron	0.5m *1	280
6	L-Angle 15x15mm	Cast iron	0.25m	140
7	Clamp	Alloy steel	2	50
8	Fevicol	NIL	100ml *	75
9	Nut ,bolt washer	Mild Steel	As required	100
10	Welding electrodes	NIL	As required	150
11	Travelling allowance	NIL	NIL	500
12	Other costs	NIL	NIL	500
TOTAL COST (In Rupees)				2616/-

Since the project is done with low cost materials and cheaper manufacturing methods are employed for producing the components like airfoil and radial arms the cost for this Giromill wind turbine is very low. But when it is decided to build a large scale Giromill wind turbine with appropriate methods and suitable materials the approximate cost for production may reach Rs. 10,000/-. The readymade vertical axis wind turbines are also available in internet for domestic purposes ranging from Rs. 10,000/- to Rs. 25,000/-. The cost of the wind turbine increases with increase in power output and reliability.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The project has been completed successfully. The project work was developed after conducting number of experiments before finalizing the design work.

In general the entire developments of the project work was educative and we could gain a lot of experience by way of doing the project practically. We could understand the practical constraints of developing such systems about which we have studied by way of lectures in the theory classes. It was satisfying to see so many theoretical aspects work before us in real life practice of which we have heard through lectures and of which we have studied in the books.

Since the vertical axis wind turbines are having low efficiency than horizontal axis wind turbines it is not commonly used in our country. But these kinds of wind turbines are highly suitable for small scale domestic purposes at low costs when compared to HAWT. Due to increasing demand for renewable energy it is hoped that these kinds of VAWT plays an important role in every home by assisting the energy needs.

7.2 FUTURE SCOPE

Apart from the design and fabrication of the project it is planned to analyze the power coefficient characteristics based on the helix angle of the airfoil. This can be done by making a small model of the wind turbine and testing it with the help of the wind tunnel.

Since the Giromill and Darrieus wind turbines are not self starting in nature, it is planned to produce a mechanism that helps the automatic pitching of the blades. So that the wind turbine acts as a drag type while starting and automatically turns into lift type when a particular speed is attained.

Since the renewable energy development section becomes a very important engineering section the project may find good developments in future.

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